

Draft - expert review pending

This unit is complete and passes the platform's automated checks, but no reviewer has read it yet. Treat its sources and claims with care.

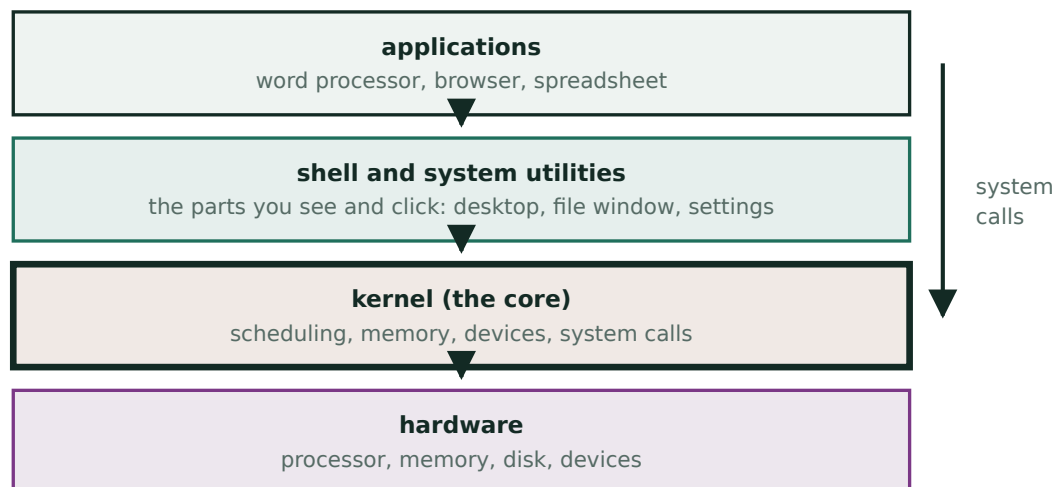
Topic 3.4 - The kernel: the core of the system

A real classroom situation

During a power fluctuation the lab's computer restarts and shows a strange half-screen message about a "kernel" before recovering. A pupil asks Mr. Waqar what a kernel is. He is honest: "I have restarted this machine a thousand times and never met the kernel. But it has been meeting me every single time." The kernel is the part of the operating system that is always there, from the first instruction after the power button to the last moment before shutdown - the layer everything else stands on.

The four layers of the system

Applications ask; the kernel decides; the hardware acts



The kernel is drawn with the thickest border: everything stands on it.
"Linux" strictly names a kernel; a full operating system bundles it with shell and utilities.

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Explanation

Kernel functions

The kernel is the core of the operating system: the first major program the machine loads after the start-up instructions, and the part that never stops running. Everything else - the desktop, the windows, the utilities, the applications - is arranged around it.

Picture the system as layers. At the bottom sits the **hardware**: processor, memory, disk, devices. At the top sit the **applications** you use. Between them run two layers of the operating system: the **shell and utilities**, which are the parts you see and click (the desktop, the file window, the settings screens), and beneath those the **kernel**, which does the fundamental work. The shell is the face; the kernel is the engine room.

The kernel's functions are the deep versions of what the earlier topics described from the outside. It **schedules** processes: it decides who runs next and for how long - the "which process should run" decision that Arpaci-Dusseau and Arpaci-Dusseau (2023) describe as the OS's policy work. It **manages memory**: it hands each process its address space and keeps processes out of one another's way, so one crashing program does not take the machine with it. It **controls devices**: every driver operates under the kernel's supervision, and all device traffic passes through it. And it exposes **system calls**: the small, well-guarded doorways through which applications request services - open a file, write to the screen, start a process - instead of touching hardware themselves.

That last point is the kernel's deepest rule: ordinary applications never touch hardware directly. They ask, through a system call, and the kernel decides whether the answer is yes. This is why the system survives a badly written program: the program can ask for anything, but the kernel is the one that acts, and it acts within the rules. It is also why the kernel failing is so serious - the half-screen message in the classroom story was the machine reporting that its engine room had a problem and needed to restart.

The kernel's four functions

Function	What it does in plain words	Where you met it
Scheduling	decides which process runs next and for how long	Topic 3.2, process states
Memory management	gives each process its address space and keeps them apart	Unit 2, RAM and the storage hierarchy
Device control	supervises all drivers and all device traffic	Topic 3.3, the input/output chain
System calls	the guarded doorways applications use to request every service	every print, save and open you have ever done

Applications never touch hardware directly: they ask, and the kernel decides.

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A note on a distinction that trips many learners: kernel and operating system are not two names for the same thing. The operating system is the whole package - kernel plus shell plus utilities. The kernel is the essential core within it. When someone says "Linux", strictly speaking they name a kernel; the familiar desktops built on it bundle that kernel with shells and utilities to make a complete operating system. In everyday speech the words blur; in an exam answer they should not.

Activity: Layer sort

Work in pairs or threes. About 15 minutes.

1. Your teacher gives you twelve cards, each naming one thing: word processor, desktop and icons, the scheduler, a printer driver, the browser, the file window, memory allocation, the disk controller's traffic, a system call to open a file, the settings screen, the power-on start-up instructions, the biometric machine's software.
2. Sort the cards into the four layers: applications, shell and utilities, kernel, hardware (the last card belongs to a device's own side of the boundary).

3. For any card your pair argues about, write the argument and settle it by asking: does it do a task for the user, draw what the user sees, do the core work, or sit in the machine itself?
4. Be ready to defend one placement each when the class compares sorts.

Check your understanding

1. Define the kernel in one sentence and state where it sits among the system's layers. (*Remember*)
2. Name the kernel's four functions from this topic. (*Remember*)
3. A word processor wants to save a file. Explain why it cannot write to the disk itself, and what happens instead. (*Apply*)
4. Explain the difference between the kernel and the operating system, using the Linux example. (*Understand*)

Summary

The kernel is the operating system's core: the always-running layer between the shell above it and the hardware below. It schedules processes, manages memory, controls devices through their drivers, and exposes the guarded doorways (system calls) through which applications must request every service. Applications never touch hardware directly, which is what keeps one bad program from sinking the machine - and why a kernel failure is the most serious failure a computer can have.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can define the kernel and place it among the four layers.
- ☐ I can name the kernel's four functions and give a plain-language example of each.
- ☐ I can explain why applications never touch hardware directly, and what a system call is.

Try this at your practicum school

Ask the lab in-charge what the school's computers show when they fail badly (the half-screen messages, the restart loops). Write two sentences: what the message names, and which layer of the system it belongs to.

Summative task

For a staff meeting, prepare a one-page explainer titled "Who is really running the computer?" Using the four layers and the kernel's functions, explain to non-technical colleagues why closing a window is not always the same as ending a program, and why the machine protects itself from any single program.

Marking guide (10 marks):

- The four layers presented accurately and in order: 3 marks.
- The kernel's functions explained in plain language: 3 marks.
- The two questions answered using system calls and scheduling rather than assertion (Analyse): 3 marks.
- Fits the audience: 1 mark.

Further reading

- Arpaci-Dusseau, R. H., & Arpaci-Dusseau, A. C. (2023). *Operating Systems: Three Easy Pieces* (version 1.10), chapter 4.
<https://pages.cs.wisc.edu/~remzi/OSTEP/>
- Bourgeois, D. T., Smith, J. L., Wang, S., & Mortati, J. (2019). *Information Systems for Business and Beyond* (2nd ed.), chapter 3. Open Textbook Library. <https://open.umn.edu/opentextbooks/textbooks/information-systems-for-business-and-beyond>
- Shelly, G. B., & Vermaat, M. E. *Discovering Computers*. Cengage Learning. (Course guide reading; print monograph, cited at bibliographic level.)

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